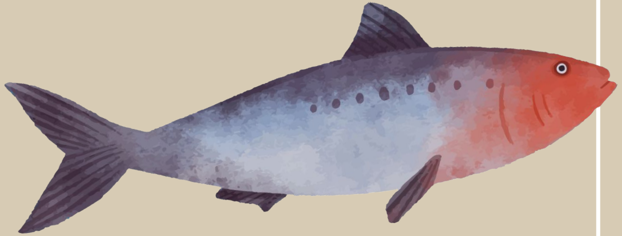
BIS 015L



Final Project

Ashlee Pilon and Maison Erridge

Quick Explanation of the Study

Fish were caught in Sarasota Bay, Florida and their tissues were examined for microplastics - tiny plastic particles under 5mm. The goal was to see if fish (prey) could be a source of plastic exposure for dolphins in the same bay.

Sample Types

Fish Muscle

The flesh of the fish, plastics here means it got into the body tissue itself

Fish GI

The gastrointestinal tract (stomach + intestines), plastics here means the fish ate it

Particle Types

Fiber_single

Smooth, uniform surface with a length that exceeded the width

Fiber_bundle

20+ fibers tangled together that can't be separated

Film

Thin flat plastic sheet, like plastic bag fragments

Fragment_non TWP

Broken piece of harder plastic, like a chunk off a bottle or container

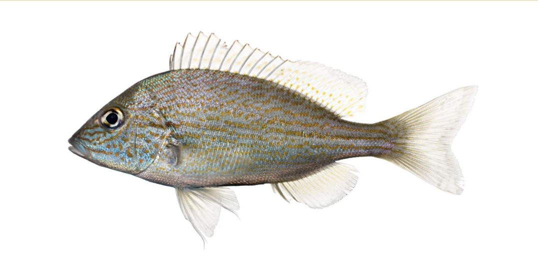
Fragment_TWP

Tire Wear Particles. Black rubber fragments that come off car tires.

Foam

Distorted when handled, but goes back to original shape

Fish Looked at in Study



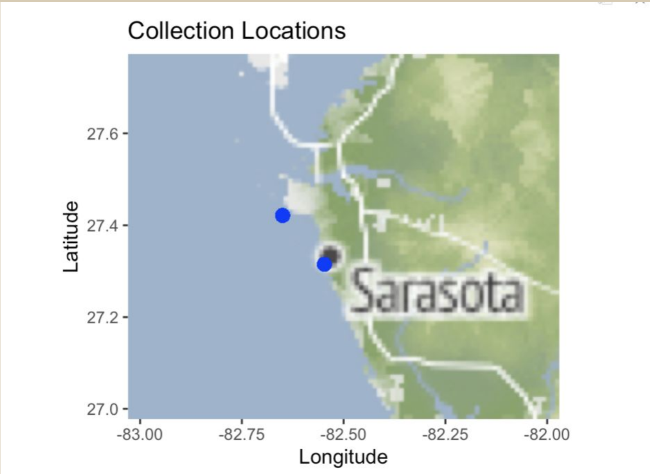
Pigfish -
Orthopristis chrysoptera

Pinfish -
Lagodon rhomboides

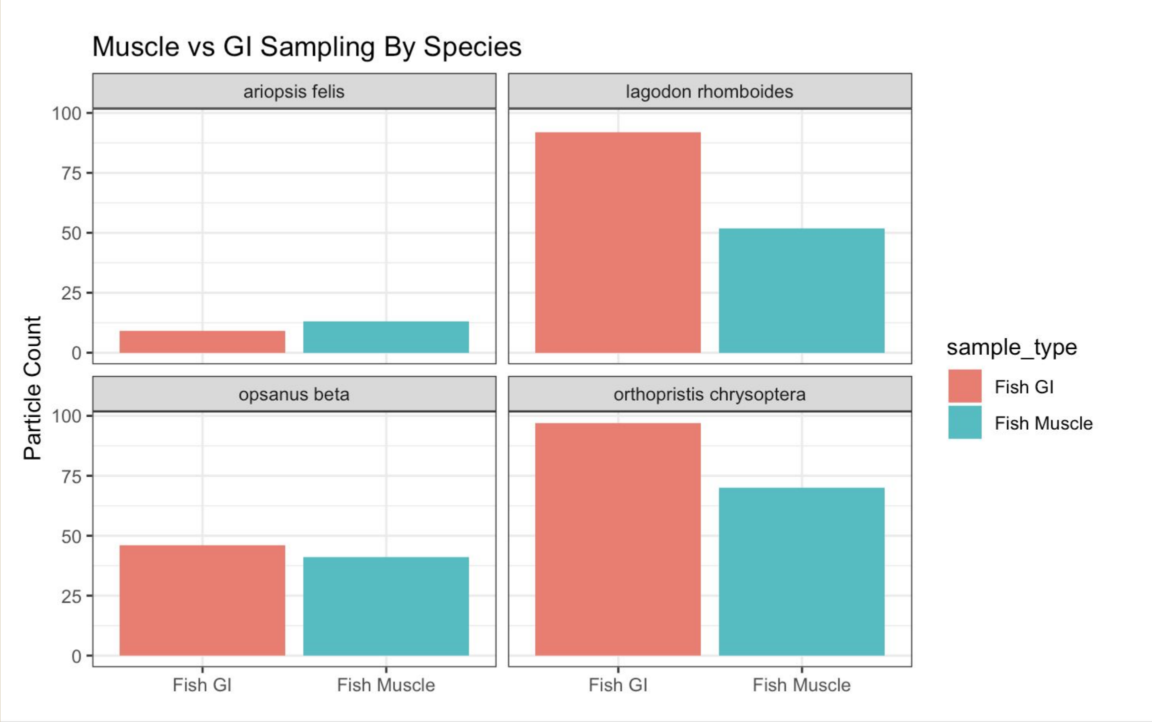
Gulf Toadfish -
Opsanus beta

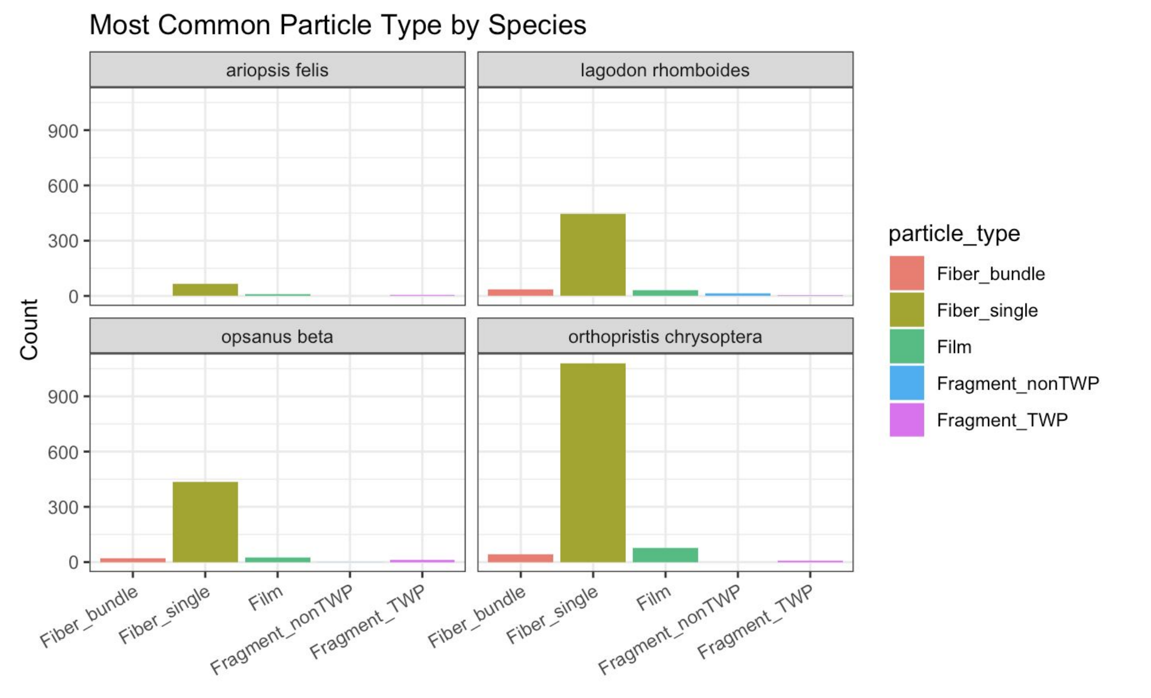
Hardhead Catfish -
Ariopsis felis

Where the Study Was Done

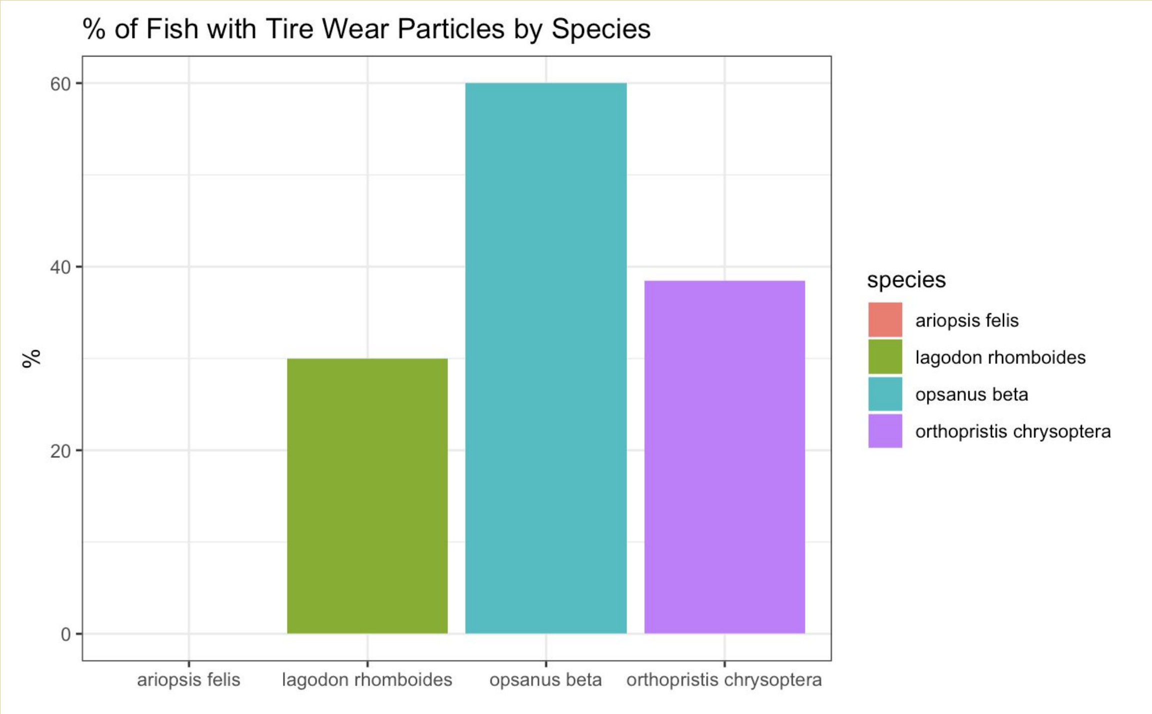


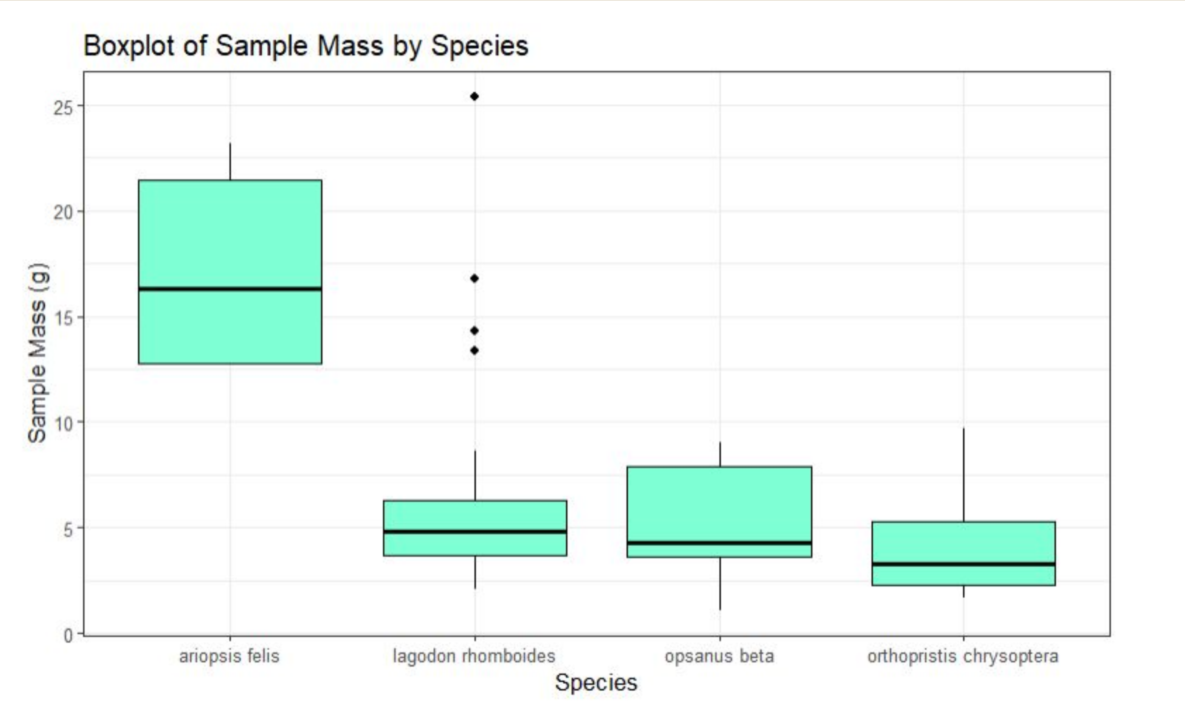
Graphs


Muscle vs GI Sampling By Species
ariopsis felis
lagodon rhomboides
opsanus beta
orthopristis chrysoptera
Particle Count
100
75
50
25
0
Fish GI
Fish Muscle
sample_type
Fish GI
Fish Muscle



Most Common Particle Type by Species
ariopsis felis
lagodon rhomboides
opsanus beta
orthopristis chrysoptera
Count
900
600
300
0
Fiber_bundle
Fiber_single
Film
Fragment_nonTWP
Fragment_TWP
particle_type
Fiber_bundle
Fiber_single
Film
Fragment_nonTWP
Fragment_TWP


% of Fish with Tire Wear Particles by Species
60
40
20
0
%
ariopsis felis
lagodon rhomboides
opsanus beta
orthopristis chrysoptera
species
ariopsis felis
lagodon rhomboides
opsanus beta
orthopristis chrysoptera


Boxplot of Sample Mass by Species
Sample Mass (g)
25
20
15
10
5
0
ariopsis felis
lagodon rhomboides
opsanus beta
orthopristis chrysoptera
Species

Shiny App

Takeaways From Data

- Microplastics were found in nearly every fish sampled, confirming that microplastic contamination is widespread in local fish populations.
- Fiber_single particles dominated every species. We hypothesize that it originates from synthetic fabrics like polyester that shed during washing.
- Tire wear particles showed up in 3 out of 4 species. 60% of toadfish contained tire rubber fragments.
- Small sample sizes creates limited conclusions. With only 2 catfish and 5 toadfish, patterns in those species should be interpreted cautiously. A larger, more balanced sample across species would strengthen these findings.

Future Research Questions

Bigger fish, more plastic?

Ariopsis felis had significantly larger sample masses than the other three species. Adding more catfish data would allow us to create a plastic to body mass ratio that would let us compare between species more effectively.

Does road runoff create microplastics?

TWP showed up in a majority of samples. Looking at samples of fish located near roads and highways could help assess if road runoff creates more microplastics.

Thank You!